

Source A

Alpha particles. ✓

Properties - It is +vely charged ✓

- low penetrating power ✓
- short range in air.
- High ionising power
- Stopped by thin sheet of paper
- Deflected toward by both Electric / Magnetic field.

Source B - Beta particle ✓

Properties - Negatively charged ✓

- Higher penetrating power than α -ples ✓
- Higher range in air than α -ples.
- lower ionising power than α -ples
- Fast moving electron
- stopped by thin aluminium foil.
- Deflected by both Electric / Magnetic field.

Source C - Gamma ray. ✓

Properties - No charge ✓

- Not deflected in both electric / Magnetic field ✓
- Highest penetrating power
- shortest range in air.
- Lowest ionising power.
- Travel faster at as speed of light.

c) 234g of Thorium contain 6.02×10^{23} molecules.

$$2g \text{ of Thorium contain } \frac{2 \times 6.02 \times 10^{23}}{234}$$

$$= 5.1453 \times 10^{21} \text{ Molecules.}$$

$$\text{Decay constant } \lambda = \frac{0.693}{t_{1/2}} \left(\frac{\ln 2}{t_{1/2}} \right)$$

$$= \frac{0.693}{5.45 \times 365 \times 24 \times 60 \times 60}$$

$$= 4.032 \times 10^{-9} \text{ s}^{-1}$$

$$\text{Activity, } A = \lambda N$$

$$= 4.032 \times 10^{-9} \times 5.1453 \times 10^{21}$$

$$= 2.0746 \times 10^{13} \text{ Bq (s}^{-1}\text{)}$$

SECTION B: FORCES AND MOTION

$$2(a) \quad v^2 = \frac{\lambda g}{2\pi} + \frac{2\pi\delta}{\lambda\rho}$$

$$\begin{aligned} [LHS] &= [v]^2 \\ &= (LT^{-1})^2 \\ &= L^2 T^{-2} \quad \checkmark \end{aligned}$$

$$\begin{aligned} [RHS] &= \frac{[\lambda][g]}{[2\pi]} + \frac{[2\pi][\delta]}{[\lambda][\rho]} \\ &= \frac{L L T^{-2}}{1} + \frac{M T^{-2}}{L M L^{-3}} \\ &= L^2 T^{-2} + L^2 T^{-2} \\ &= L^2 T^{-2} \quad \checkmark \end{aligned}$$

$$\therefore [LHS] = [RHS]$$

(02)

Dimensions of a physical quantity are the powers to which the fundamental quantities must be raised. $\checkmark$ (01)

$$(b) \quad P \propto F^x v^y$$

$$P = k F^x v^y \quad \checkmark$$

$$[LHS] = [RHS]$$

$$[P] = [k][F]^x [v]^y$$

$$M L^2 T^{-3} = (M L T^{-2})^x (L T^{-1})^y$$

$$M L^2 T^{-3} = M^x L^{x+y} T^{-2x-y} \quad \checkmark$$

$$x = 1 \quad \checkmark \quad \text{--- (1)}$$

$$2 = x + y \quad \text{--- (2)}$$

$$-3 = -2x - y \quad \text{--- (3)}$$

From (2) and (1)

$$2 = 1 + y$$

$$y = 1 \quad \checkmark$$

From (3) and (1)

$$-3 = -2(1) - y$$

$$-3 = -2 - y$$

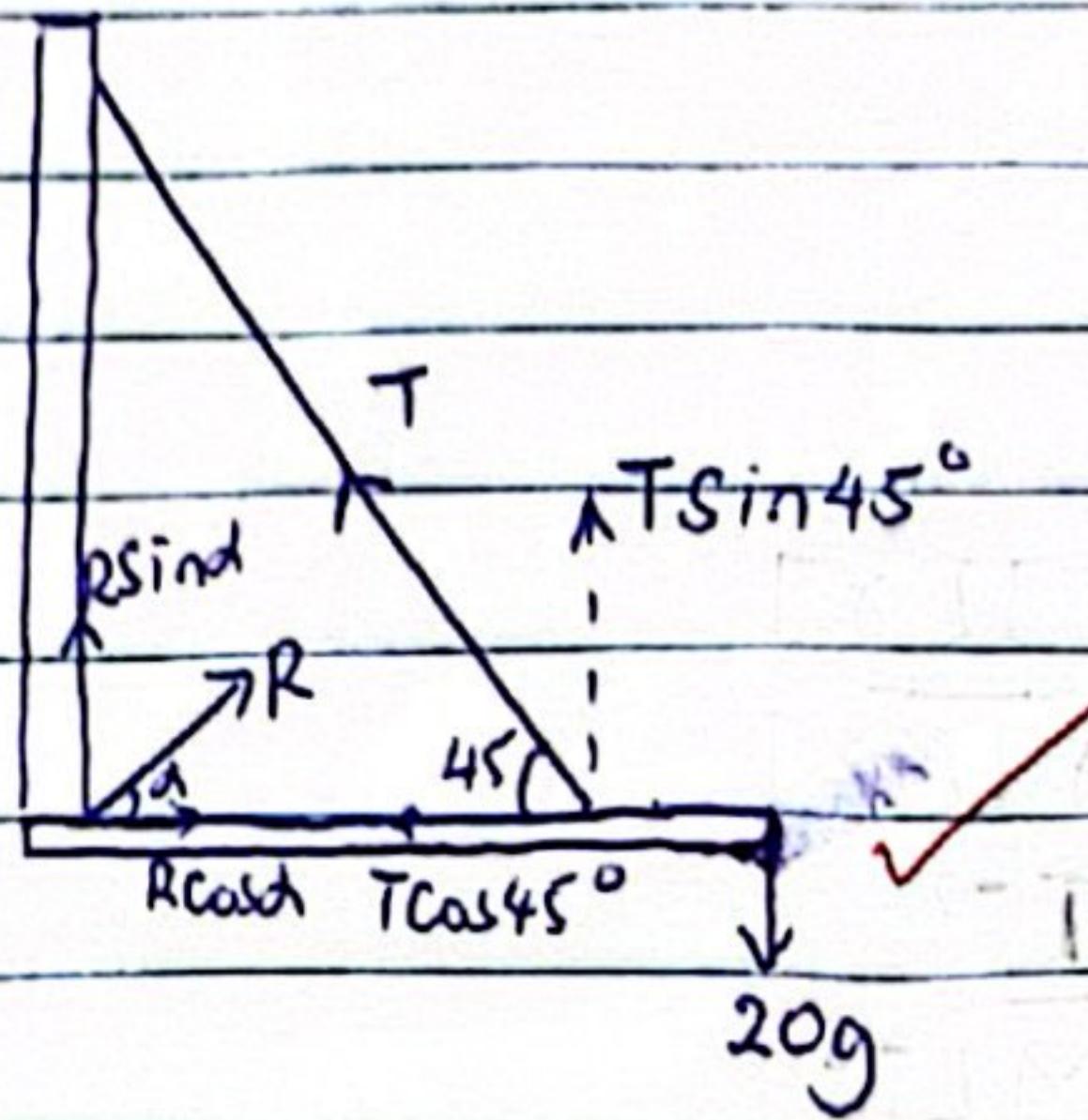
$$y = 1$$

$$\therefore P = k F v \quad \checkmark$$

(05)

- (c) When a body is in mechanical equilibrium, the sum of clockwise moments about any point is equal to the sum of anticlockwise moments about the same point ✓ (01)

d. i)



Taking moments about the hinge,

$$20g \times 2 = (T \sin 45^\circ)(1.5) \quad \checkmark$$

$$T = 369.9583 \text{ N} \quad \checkmark$$

(03)

ii)

$$R \cos \alpha = T \cos 45^\circ$$

$$R \cos \alpha = 369.9583 \cos 45^\circ$$

$$R \cos \alpha = 261.6 \quad \text{--- (1) } \checkmark$$

$$R \sin \alpha + T \sin 45^\circ = 20g \quad \checkmark$$

$$R \sin \alpha = 20g - T \sin 45^\circ$$

$$R \sin \alpha = 20g - 369.9583 \sin 45^\circ$$

$$R \sin \alpha = -65.4 \quad \text{--- (2) } \checkmark$$

$$(2)^2 + (1)^2;$$

$$R^2 = 261.6^2 + (-65.4)^2$$

$$R = 269.6511 \text{ N} \quad \checkmark$$

$$(2) \div (1)$$

$$\tan \alpha = \frac{-65.4}{261.6}$$

$$\alpha = -14.0^\circ \quad \checkmark$$

$\therefore$ The reaction at the hinge is 269.6511 N at an angle of -14.0° to the horizontal. ✓ (06)

iii) Let the mass be m

Taking moments at the hinge with $T = 500 \text{ N}$

$$500 \sin 45^\circ \times 1.5 = (20 + m)g \times 2 \quad \checkmark$$

$$m = 7.0301 \text{ kg} \quad \checkmark$$

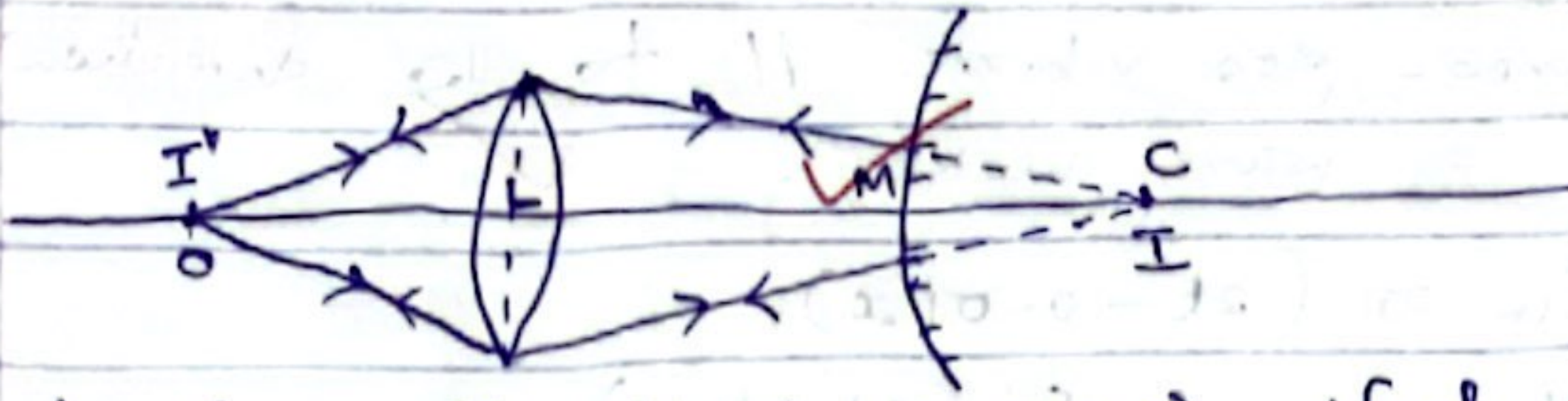
(02)

The maximum additional mass is 7.0301 kg.

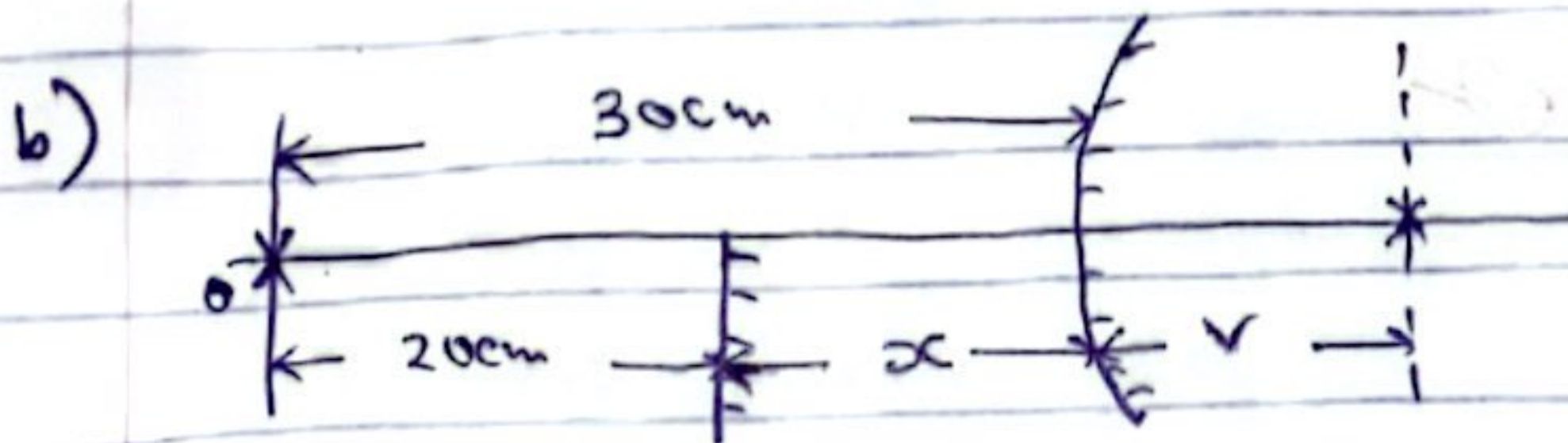
$u = 1.56$
 7.3902

Item 2

- a) To determine the focal length of a convex mirror using a convex lens.



- An illuminated object is placed in front of a convex lens at a distance greater than the focal length of the lens.
- A screen is placed behind the lens and adjusted until a sharp image I is formed on the screen.
- The distance LI is measured. ✓
- A convex mirror whose focal length is to be determined is placed b/w the lens and the image with its reflecting surface facing the lens. ✓
- The mirror is adjusted until the ~~final image~~ coincides with the object O. ✓
- The rays from the lens are ~~then~~ incident on the mirror normally, C is the centre of curvature of the mirror.
- The distance LM is measured and recorded. ✓
- The radius of curvature $r = -MC = -(LI - LM)$ ✓
- The focal length f is given by $f = \frac{r}{2}$ ✓



$$v = -(20 - x) \text{ but } x = 30 - 20 = 10\text{cm}$$

$$v = -(20 - 10)$$

$$= -10\text{cm} \quad \checkmark$$

Using $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ ✓

$$\frac{1}{f} = \frac{1}{30} + \frac{1}{-10} \quad \checkmark$$

$$f = -15\text{cm} \quad \checkmark$$

The focal length of the convex mirror is 15cm

c) Transverse waves - particles vibrate perp to the direction of the wave motion. 02

Longitudinal waves - particles vibrate // to the direction of the wave motion. 02

Given $y = 0.01 \sin 2\pi (2t - 0.01x)$

Compare with $y = A \sin(\omega t - kx)$

$A = 0.01 \text{ m}; \omega = (2\pi)2 = 4\pi; k = 2\pi(0.01)$

(i) $\omega = 2\pi f$ 02

$4\pi = 2\pi f$

$f = 2 \text{ Hz}$ 02

(ii) $k = \frac{2\pi}{\lambda}$

$0.02\pi = \frac{2\pi}{\lambda}$

$\lambda = 100 \text{ m}$ 04

$v = \lambda f$

$v = 2 \times 100$

$v = 200 \text{ m/s}$ 02

(iii) Phase difference

$\phi = \frac{2\pi x}{\lambda}$ 02

$= \frac{2\pi(50)}{100}$

$= \pi$ 02



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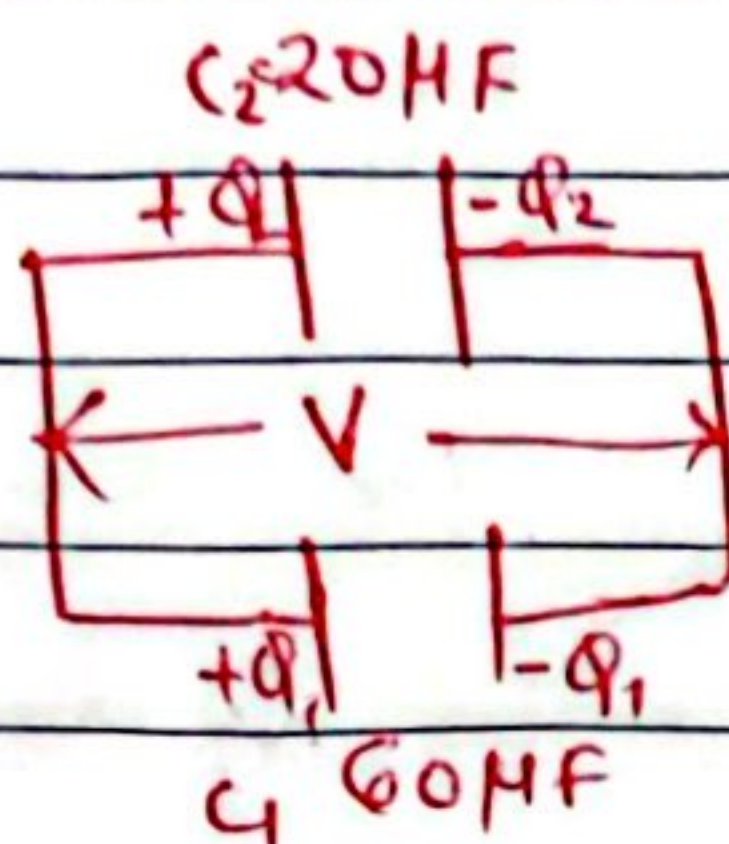
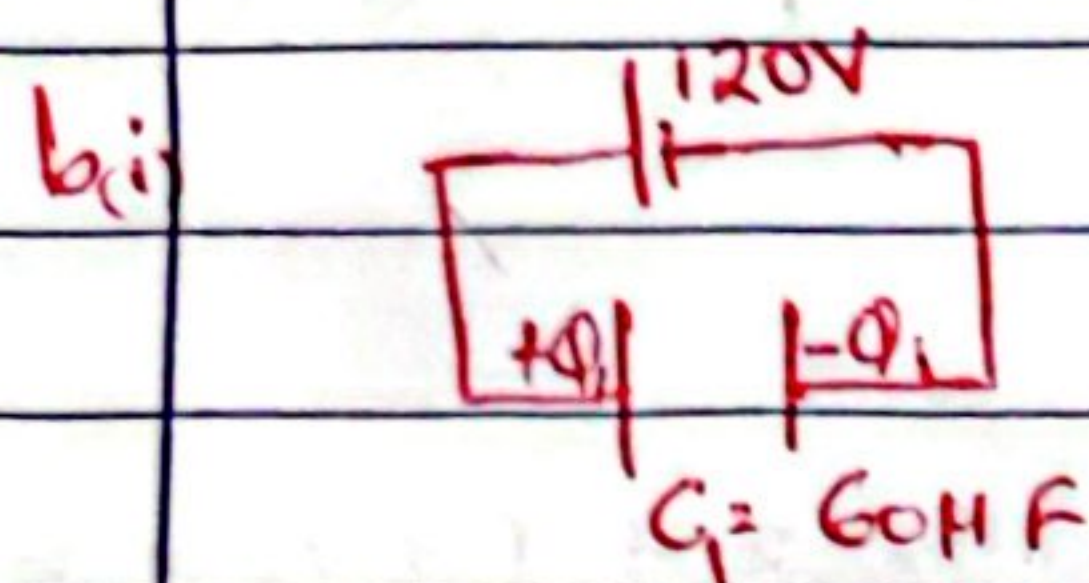
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a) $C = \frac{Q}{V}$; farads (F) ✓ 02



Total charge, $Q = CV$ ✓
 $= 60 \times 10^{-6} \times 120$
 $= 7.2 \times 10^{-3} C$ ✓
 or $0.0072 C$ before combination.

After combination

$Q = Q_1 + Q_2$
 $= C_1 V + C_2 V$
 $= (C_1 + C_2) V$
 $= (60 + 20) \times 10^{-6} V$ ✓
 $= 0.00008 V$ or $8.0 \times 10^{-5} V$ ✓

0.5

By conservation of charge,

Total charge before combination = Total charge after combination
 $0.0072 = 0.00008 V$ ✓
 $V = 90 V$ ✓

∴ The final p.d after the capacitors are connected is 90V.

ii) Energy stored, $E = \frac{1}{2} CV^2$ ✓

Initial stored energy, $E_1 = \frac{1}{2} \times (60 \times 10^{-6}) \times 120^2$ ✓
 $= 0.432 J$ ✓

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final stored energy, $E_2 = \frac{1}{2} (C_1 + C_2) V^2$

$$= \frac{1}{2} (60 + 20) \times 10^{-6} \times 90^2$$

$$= 0.324 \text{ J} \quad \checkmark$$

Energy difference = $E_1 - E_2$

$$= 0.432 - 0.324$$

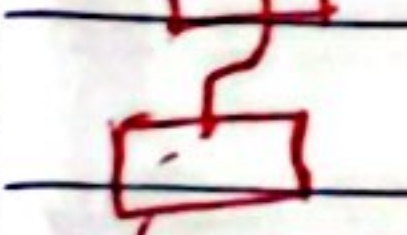
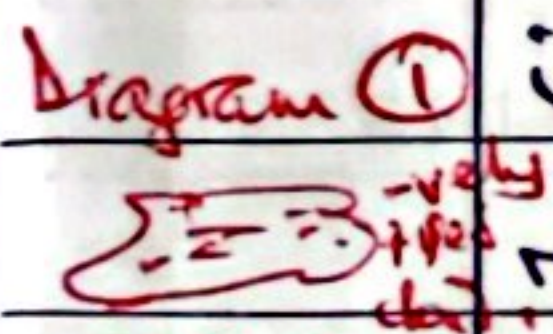
$$= 0.108 \text{ J} \quad \checkmark$$

Energy difference observed is due to some energy lost as heat in the connecting wires when the capacitors are joined. (05)

c) Action of a lightning conductor. (4)

d) Faraday's ice pail expt (4)

c) When a -vely charged cloud passes over a lightning conductor, it induces +ve charges on the spikes and -ve charges are repelled to the ground through the copper strip.



Copper plate

- Due to high charge density and electric field intensity, the air around gets ionized causing +ve and -ve ions.

- The -ve ions are attracted to and discharged at the spikes while the +ve ions are repelled to form a space charge which neutralizes the charge on the cloud.

This reduces the harmful effect of the cloud.



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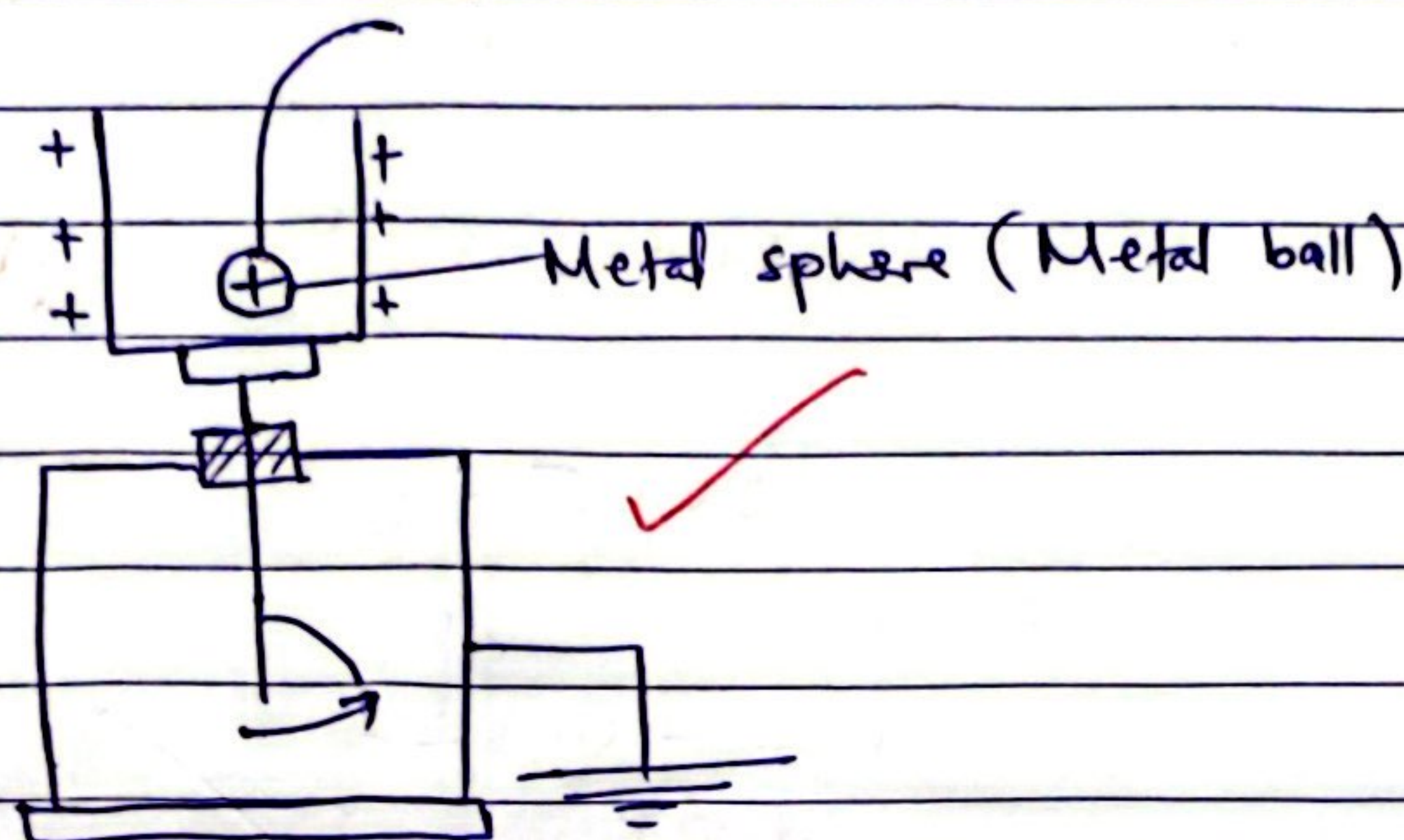
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- A ~~tr~~vely charged metal ball is lowered into the ice pail without touching the sides of the pail.
- +ve charge is induced on the outside of the pail and the leaf diverges.
- The metal ball is then allowed to touch the bottom of the pail. The divergence of the leaf does not change.
- The ball is then removed and tested for charge and it is found to be neutral.

Conclusion:

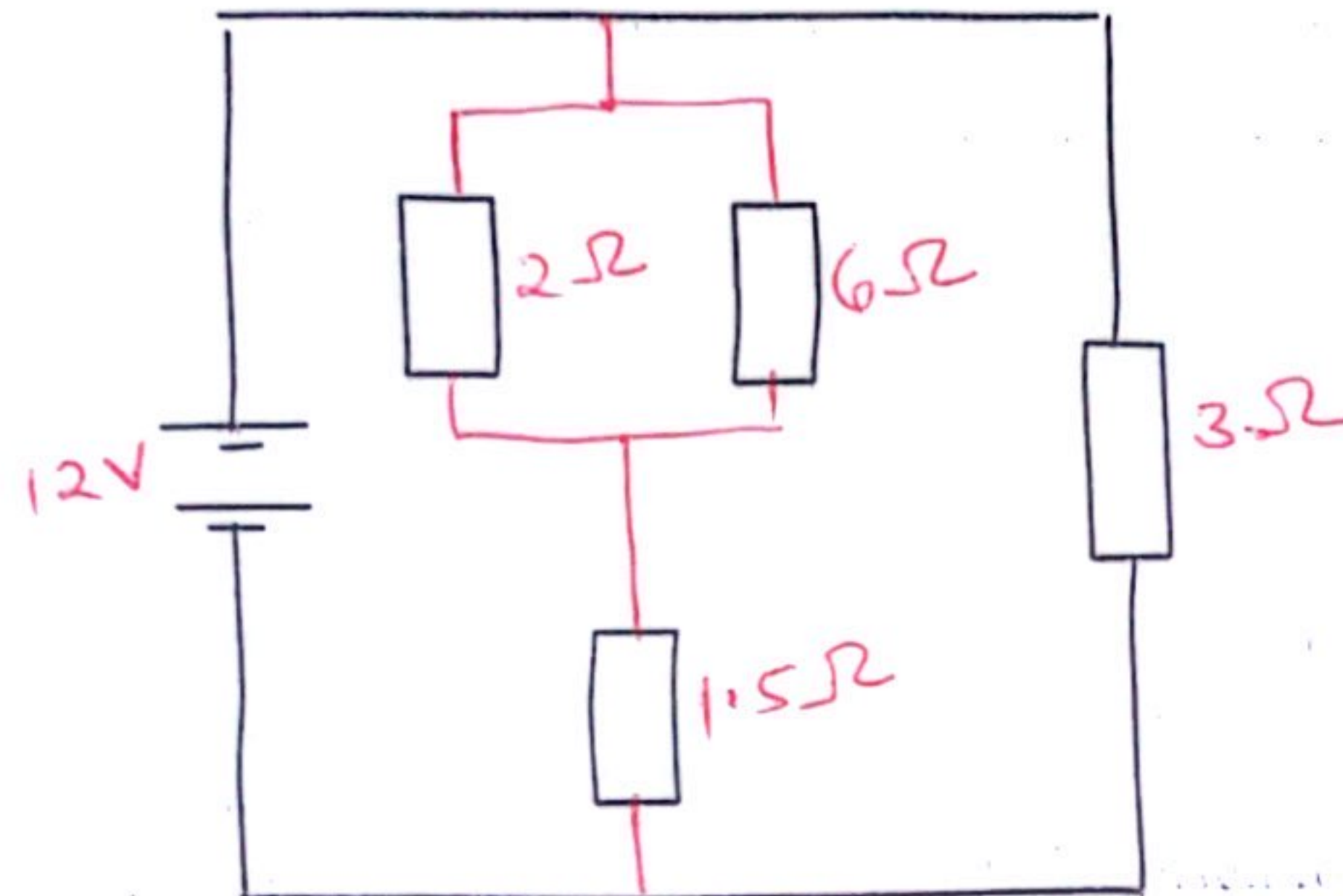
The total charge inside the ~~conduct~~ pail (hollow conductor) is zero thus excess charge resides on the outside. Since the leaf remained diverged.

64

ITEM 5

In a workshop in Kampala, an electrician is installing a low-voltage lighting system powered by a 12V d.c battery of negligible internal resistance. He connects resistors of $2\ \Omega$, $6\ \Omega$, $1.5\ \Omega$, and $3\ \Omega$ in a network as shown in the setup.

While testing the system, a student observes that the materials used for wiring affect how current flows and recalls that some conductors obey a linear current voltage relationship. To obtain accurate measurements of resistance, the student later uses a long uniform resistance wire connected to a power source, with a sliding contact used to balance potential differences along the wire, in order to determine the resistance of a sample wire.



TASK

- (a) State the law that relates current and potential difference, identify one material in such wiring that follows this behavior and describe its current voltage characteristic.
- (b) Using the resistor values and d.c supply provided, calculate the power dissipated in the $3\ \Omega$ resistor used in the setup.
- (c) Explain how the student can determine the resistance of the wire.

END

TASK

- (a) State the law that relates current and potential difference, identify one material in such wiring that follows this behavior and describe its current voltage characteristic.

(b) Using the resistor values and d.c supply provided, calculate the power dissipated in the $3\ \Omega$ resistor used in the setup.

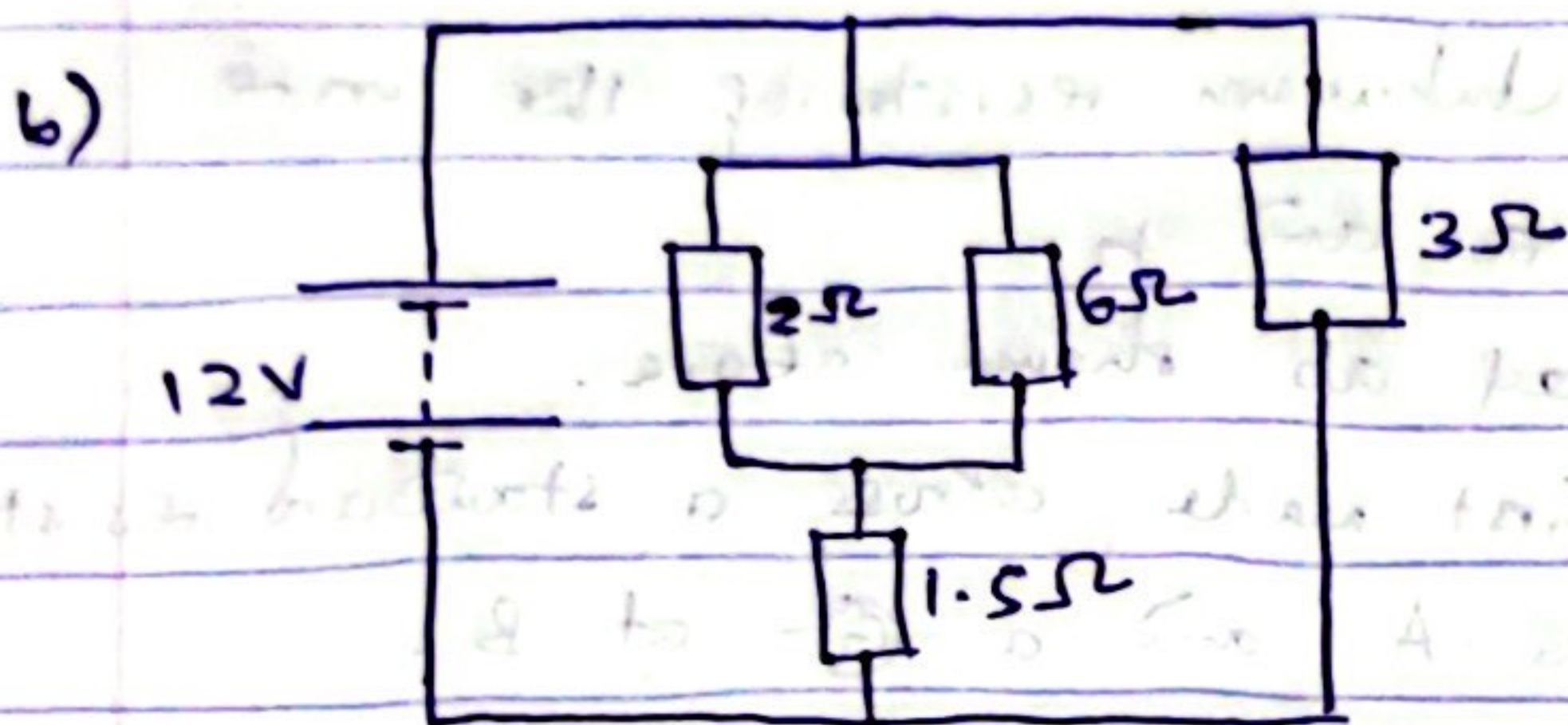
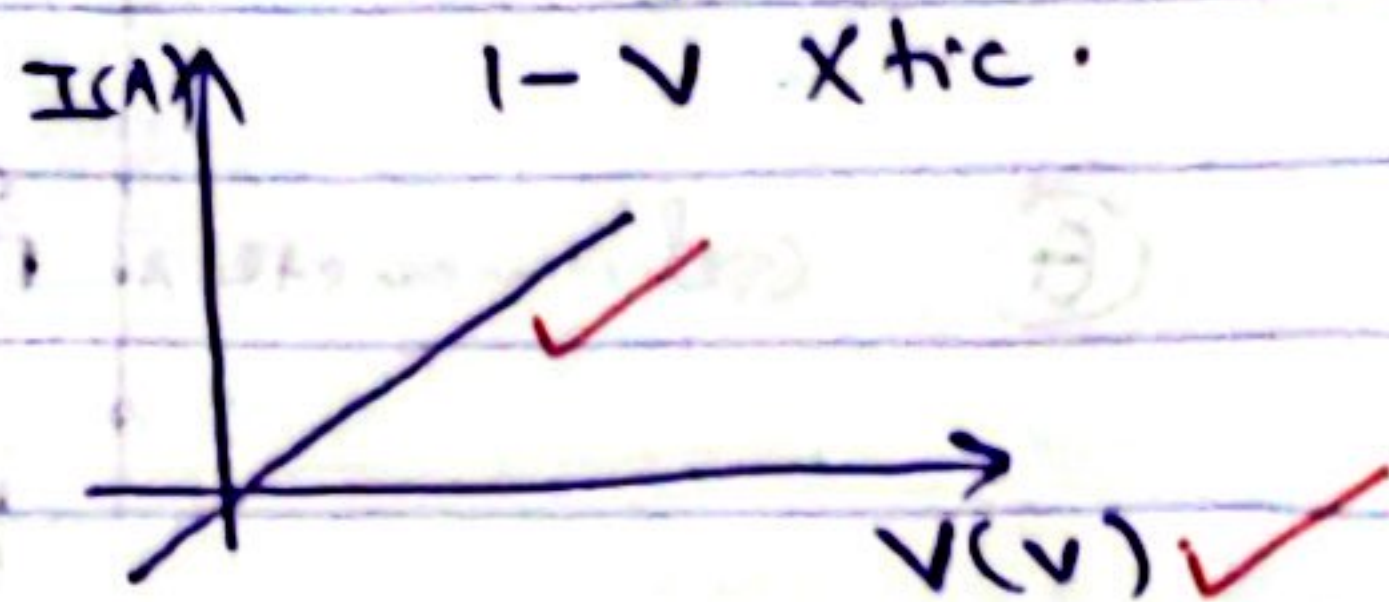
(c) Explain how the student can determine the resistance of the wire.

(d) Explain how the student can determine the resistance of the wire.

(e) Explain how the student can determine the resistance of the wire.

Item 5.

- a) Ohm's law states that the current flowing through a given conductor is directly proportional to the p.d b/w its ends provided the physical conditions remain constant.
Material - Copper, tungsten, Aluminium, Nichrome.



2Ω & 6Ω are in parallel.

$$R_p = \frac{2 \times 6}{2 + 6} = 1.5\Omega$$

$$R_s = 1.5 + 1.5 = 3\Omega$$

$$R = \frac{3 \times 3}{3 + 3} = 1.5\Omega \text{ (effective resistance)}$$

$$I = \frac{V}{R} = \frac{12}{1.5} = 8A$$

Current through 3Ω

$$I = \frac{3}{3+3} \times 8$$

$$= 4A$$

Power dissipated in 3Ω resistor

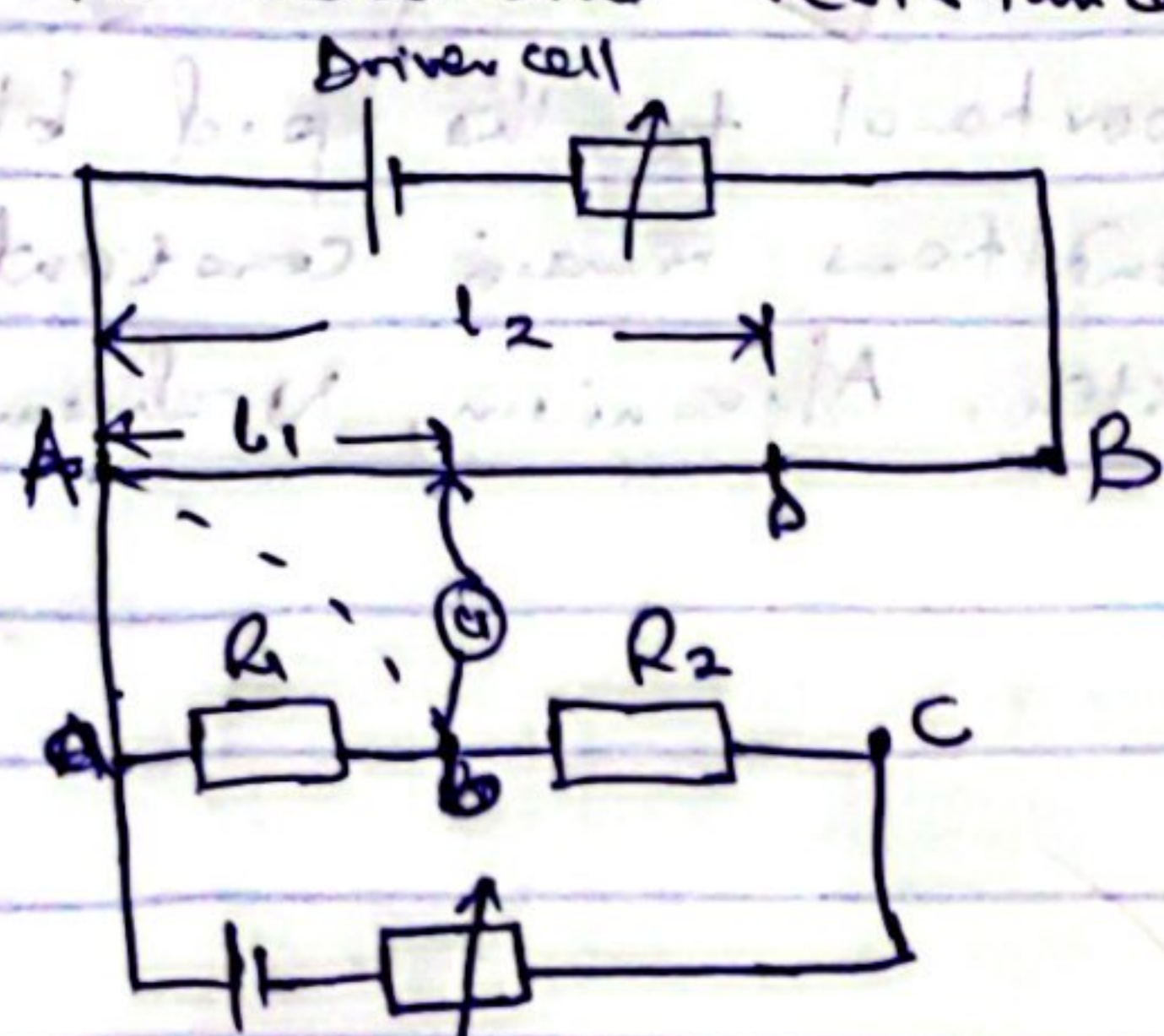
$$P = I^2 R$$

$$= 4^2 \times 3$$

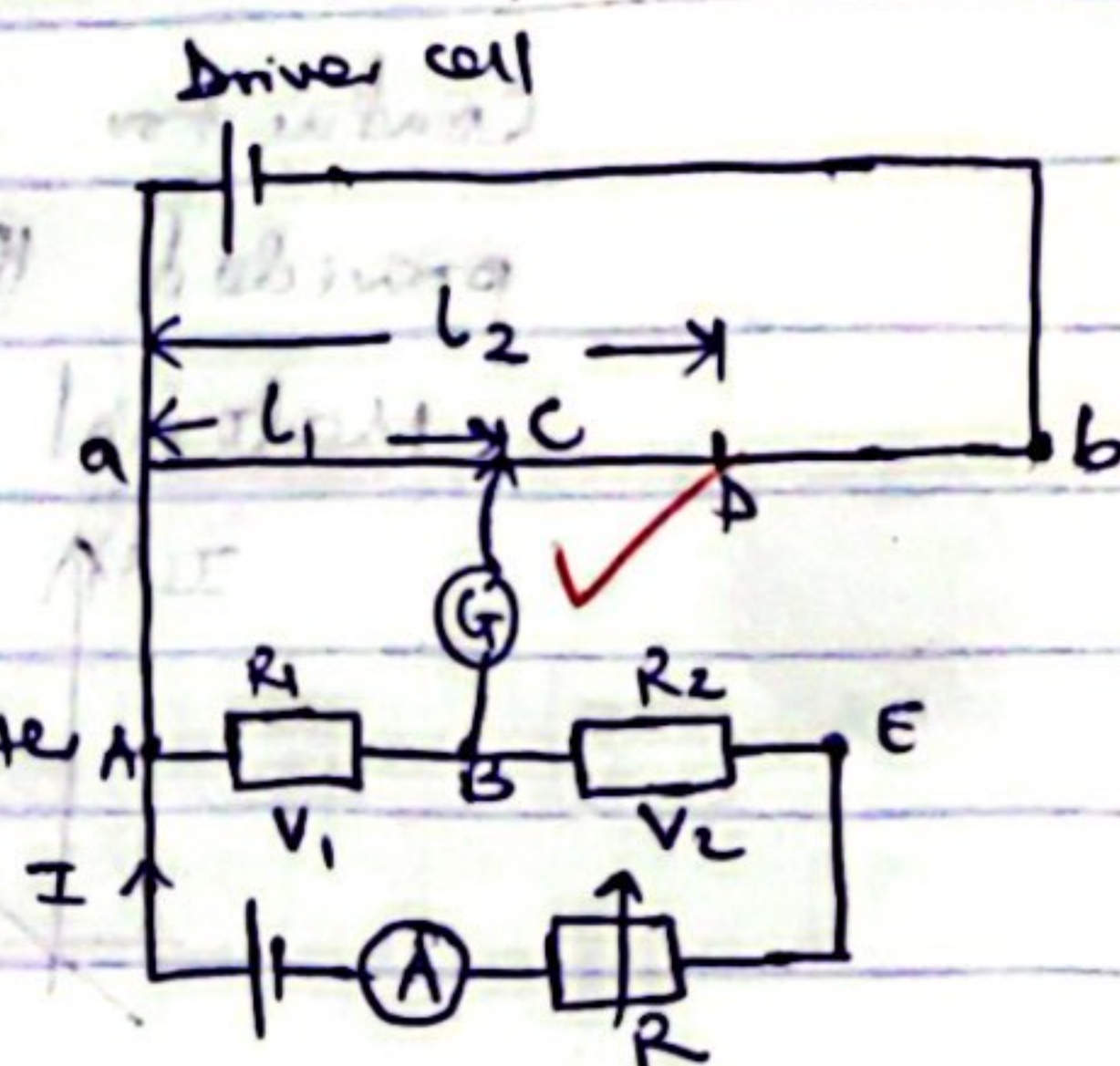
$$= 48W$$

or $P = \frac{V^2}{R} = \frac{12^2}{3}$; $12V$ is the p.d across the parallel network.
 $= 48W$

b) To determine resistance using a Potentiometer.



(G) - Galvanometer



R_1 - standard resistor

R_2 - Unknown resistor of the wire

- With connections at a and b.
- The circuit is connected as shown above.
- The connections are first made across a standard resistor R_1 by connecting point a to A and a (G) at B.
- The rheostat R is adjusted such that the ammeter records a suitable current I.
- The p.d. V_1 across R_1 is balanced along the potentiometer wire ab until a balance point C is got at which the (G) shows no deflection.
- The balance length l_1 is noted.
- Connections across R_1 are removed and they are made across an unknown resistor R_2 by connecting the (G) at E.
- The p.d. V_2 across R_2 is balanced along ab until a balance point D is got at which the (G) shows no deflection.
- The balance length l_2 is noted.
- The resistance of the wire R_2 is obtained from

$$R_2 = \left(\frac{l_2}{l_1} \right) R_1$$